Familial Juvenile Hyperuricemic Nephropathy Type 1 (ADTKD-UMOD): A Comprehensive Disease Characteristics Report

Disease: Familial Juvenile Hyperuricemic Nephropathy Type 1 **MONDO ID:** MONDO:0008073 · **OMIM:** 162000 · **Causal gene: UMOD** (uromodulin / Tamm-Horsfall protein) **Current classification:** Autosomal Dominant Tubulointerstitial Kidney Disease, UMOD subtype (**ADTKD-UMOD**)

Summary

Familial Juvenile Hyperuricemic Nephropathy Type 1 (FJHN1; MONDO:0008073; OMIM 162000) is caused by heterozygous, mostly missense, gain-of-toxic-function mutations in *UMOD*, the gene on chromosome 16p12.3 encoding uromodulin (Tamm-Horsfall protein). The causal gene is **UMOD** — it is important to state this explicitly because the historical name "familial juvenile hyperuricemic nephropathy" was once linked in older literature to a putative *SLC8A1* locus by linkage analysis, but the established molecular cause of Type 1 disease is UMOD. Under the 2015 KDIGO consensus, FJHN1, medullary cystic kidney disease type 2 (MCKD2), uromodulin-associated kidney disease (UAKD), and glomerulocystic kidney disease with UMOD mutation were unified into a single entity: **ADTKD-UMOD**. This disease is distinct from similarly named entities caused by other genes (*MUC1*, *REN*, *HNF1B*, *SEC61A1*, *DNAJB11*); the identifiers above anchor this report.

Mechanistically, pathogenic UMOD variants — predominantly affecting conserved cysteine residues in the N-terminal EGF-like and cysteine-rich/D8C domains — cause misfolding and retention of mutant uromodulin in the endoplasmic reticulum (ER) of thick ascending limb (TAL) epithelial cells. ER retention triggers the unfolded protein response (specifically the IRE1/XBP1 branch), ER stress, disrupted proteostasis and mitochondrial dynamics, defective autophagy, and progressive tubular cell death with interstitial fibrosis. This produces slowly

progressive chronic kidney disease (CKD) that reaches end-stage kidney disease (ESKD) typically between the third and seventh decades. In parallel, TAL dysfunction impairs NKCC2-mediated salt handling, triggering compensatory proximal-tubule sodium-coupled urate reabsorption (URAT1/SLC22A12, SLC5A8), which reduces fractional excretion of uric acid and produces the characteristic **early-onset hyperuricemia and gout** (often in the teenage years).

ADTKD-UMOD is one of the more common monogenic causes of CKD (ADTKD collectively ~5% of monogenic CKD; UMOD the most prevalent ADTKD gene), yet it remains substantially under-diagnosed because it presents with bland urine (little/no proteinuria or hematuria) and non-specific CKD. Diagnosis rests on autosomal-dominant family history, hyperuricemia/gout disproportionate to kidney function, and confirmatory *UMOD* sequencing; characteristic histology shows tubulointerstitial fibrosis with intracytoplasmic uromodulin aggregates. There is **no proven disease-modifying therapy**: management is supportive (xanthine oxidase inhibitors such as allopurinol/febuxostat for gout, nephroprotection, and curative kidney transplantation without graft recurrence). Autophagy-enhancing proteostasis strategies (e.g., calorie restriction) show promise in mouse models and represent the leading experimental therapeutic direction.

Key Findings

Finding 1 — FJHN Type 1 is ADTKD-UMOD, caused by gain-of-toxic-function *UMOD* mutations

Familial juvenile hyperuricemic nephropathy type 1 (MONDO:0008073) is caused by heterozygous mutations in **UMOD** (chromosome 16p12.3), encoding uromodulin/Tamm-Horsfall protein. The 2015 KDIGO consensus report reclassified FJHN, MCKD type 2, and uromodulin-associated kidney disease (UAKD) into one entity, **ADTKD-UMOD**. The disease is defined by a classic clinical triad: autosomal-dominant inheritance, progressive renal failure beginning in the third to fifth decade, and early-onset gout/hyperuricemia. Most pathogenic UMOD variants are missense substitutions affecting cysteine residues in the N-terminal cysteine-rich, EGF-like, and D8C domains (e.g., Arg185His, Cys223Gly, Cys255Tyr, Trp258Gly, Cys147Trp). The unifying molecular consequence is a **gain-of-toxic-function** caused by retention of misfolded uromodulin in the ER.

> *"Rare autosomal dominant tubulointerstitial kidney disease is caused by mutations in the genes encoding uromodulin (UMOD), hepatocyte nuclear factor-1β (HNF1B), renin (REN), and mucin-1 (MUC1)"* — KDIGO consensus PMID: 25738250
>
> *"The mutation has been traced to the UMOD gene in chromosome 16. The disease is characterised by the classic triad of autosomal dominant inheritance, progressive renal failure beginning in the third to fifth decade of life and gout."* PMID: 26872483

Finding 2 — Mutant uromodulin ER retention triggers the unfolded protein response and ER stress

In immortalized mouse TAL cells expressing a mutant GFP-uromodulin, 109 genes were differentially expressed relative to wild type, with up-regulation of ER-resident chaperones and protein disulfide isomerases. Mutant uromodulin induced the **unfolded protein response (UPR)**, specifically the IRE1 branch, evidenced by increased splicing of XBP1. The common primary effect of UMOD mutations is ER retention; downstream consequences include ER stress, altered protein and mitochondrial homeostasis, defective autophagy, and increased cell death.

> *"mutant uromodulin expression induces the Unfolded Protein Response (UPR), and specifically the IRE1 branch, as shown by an increased splicing of XBP1"* PMID: 28437467
>
> *"UMOD mutations have a common gain-of-toxic-function effect, causing mutant uromodulin retention in the endoplasmic reticulum (ER). This leads to ER stress, alteration of protein homeostasis and mitochondrial dynamics, defective autophagy, and increased cell death."* PMID: 41632531

Finding 3 — Hyperuricemia arises from impaired TAL salt handling driving compensatory urate retention

Uromodulin is produced exclusively in the TAL and regulates the NKCC2 cotransporter. Tamm-Horsfall protein/uromodulin-knockout mice display salt and water wasting via decreased luminal NKCC2 translocation, with **compensatory up-regulation of proximal-tubule sodium-coupled urate transporters SLC5A8 and SLC22A12 (URAT1)** and NHE3, elevated renin,

hypertension, and elevated serum uric acid and allantoin. In Umod knock-in mouse models (C93F, A227T), the uromodulin maturation defect reduces fractional excretion of uric acid and impairs urinary concentrating ability — the direct mechanistic basis of hyperuricemia.

> *"they had significant upregulation of Na-coupled urate transporters Slc5a8 and Slc22a12 as well as sodium-hydrogen exchanger 3 (NHE3) in the proximal tubule and elevated serum uric acid and allantoin"* PMID: 29357410
>
> *"reduced fractional excretion of uric acid and nephropathological alterations including uromodulin retention in TALH cells, interstitial fibrosis and inflammatory cell infiltrations"* PMID: 23748428

Finding 4 — Diagnostic histopathology and emerging proteostasis therapy

Renal histology shows chronic tubulointerstitial nephropathy with fibrillar/intracytoplasmic uromodulin inclusions in distal tubule cells; electron microscopy reveals bundled and cystic ER; immunohistochemistry confirms intracytoplasmic uromodulin aggregates. Genetic analysis (UMOD sequencing) is the diagnostic gold standard. Historically, allopurinol has been used for hyperuricemia and gout, but its efficacy for slowing renal progression rests only on small uncontrolled studies. In Tg-Umod^C147W mice, **30% calorie restriction stimulated autophagy and degradation of mutant uromodulin**, reverting the ADTKD-UMOD phenotype at an early stage and delaying progression at an advanced stage.

> *"Immunohistological analysis confirmed intracytoplasmic aggregates of uromodulin in the distal tubules"* PMID: 31422399
>
> *"calorie restriction largely reverted autosomal dominant tubulointerstitial kidney disease (ADTKD)-UMOD phenotype, preventing inflammation, fibrosis, and kidney function decline"* PMID: 41632531

Finding 5 — Epidemiology: a leading but under-recognized monogenic cause of CKD

ADTKD collectively accounts for **~5% of monogenic disorders causing CKD**, making it one of the most common monogenic kidney diseases after autosomal dominant polycystic kidney disease (ADPKD); UMOD is the most prevalent ADTKD gene. Among adults with CKD, diagnostic variants are found in roughly 22% across studies, and UMOD ranks among the 8 genes

accounting for more than 60% of such variants. No precise population prevalence is established and the disease is considered under-diagnosed. Inheritance is autosomal dominant with high but age-dependent penetrance; males and females are affected equally.

> *"recent data suggesting that ADTKD is one of the more common monogenic kidney diseases after autosomal dominant polycystic kidney disease, accounting for ~5% of monogenic disorders causing chronic kidney disease"* PMID: 31488840
>
> *"More than 60% of these variants affected 8 genes (PKD1, PKD2, COL4A3, COL4A4, COL4A5, UMOD, MUC1, HNF1B)"* PMID: 38958599

Finding 6 — Clinical phenotype spectrum: hyperuricemia/gout, bland urine, progressive CKD to ESKD

ADTKD-UMOD presents with early-onset hyperuricemia (sometimes in childhood) due to profound renal uric-acid under-excretion, gout often in the teenage years, bland urine sediment (no or minimal proteinuria/hematuria), normal or small kidneys, impaired urinary concentrating ability, and slowly progressive CKD reaching ESKD between the third and seventh decades. UMOD mutations cause a marked decrease in uromodulin synthesis plus accumulation of abnormal uromodulin leading to tubular cell death. In a Chinese case series, median age of symptom onset was ~26.5 years and median ESKD age ~41.9 years (without renal replacement). Renal cysts (especially corticomedullary) may occur. Overlapping historical entities include MCKD type 2, FJHN, and glomerulocystic kidney disease.

> *"UMOD gene mutations are responsible for the autosomal dominant inheritance of chronic interstitial disease, leading to the need for renal replacement in the third through seventh decades of life. Individuals with UMOD mutations also suffer from hyperuricemia in childhood, and often suffer from gout in their teenage years."* PMID: 21071970
>
> *"hyperuricemia secondary to profound renal uric acid underexcretion, gout, and chronic renal disease"* PMID: 25417683

Finding 7 — Uromodulin structure: a GPI-anchored ZP-module polymer; most mutations disrupt conserved cysteines

Uromodulin (UniProt **P07911**; HGNC:12559; 640-aa precursor) is a glycosylphosphatidylinositol (GPI)-anchored glycoprotein of the TAL apical membrane, built from EGF-like domains (including a calcium-binding EGF), a cysteine-rich/D8C domain, and a bipartite zona pellucida (ZP) module (ZPN+ZPC) that drives polymerization. Cryo-EM shows uromodulin forms a one-start helical homopolymer with ~180° twist between subunits; assembly is triggered by proteolytic release of a C-terminal propeptide that otherwise anchors assembly-incompetent precursors to the membrane. Most pathogenic ADTKD-UMOD variants are missense substitutions affecting conserved cysteines (e.g., C147W, C223G, C255Y), disrupting disulfide bonds and causing a maturation/folding defect. These variants are germline, near-absent in gnomAD, and classified **pathogenic/likely pathogenic** under ACMG/AMP.

> *"The unique filament architecture suggests an assembly mechanism in which subunit incorporation could be synchronized with proteolytic cleavage of the C-terminal pro-peptide that anchors assembly-incompetent UMOD precursors to the membrane."* PMID: 32815518
>
> *"UMOD forms a one-start helix with an unprecedented 180-degree twist between subunits enfolded by interdomain linkers that have completely reorganized as a result of propeptide dissociation."* PMID: 33196145

Finding 8 — Management: xanthine-oxidase inhibitors, renal replacement therapy, and experimental autophagy enhancers

No disease-modifying therapy is proven to slow renal decline. Management is supportive: xanthine oxidase inhibitors — allopurinol (CHEBI:40279; NCIT C614) and febuxostat — control hyperuricemia and prevent gout, alongside anti-inflammatory prophylaxis (colchicine). Evidence that urate-lowering slows CKD progression is weak (small uncontrolled studies). ESKD is treated with dialysis and kidney transplantation; because the defect is intrinsic to native TAL cells, the disease **does not recur in the allograft**, making transplantation highly effective. Precipitating/nephrotoxic agents (e.g., fenofibrate, volume depletion, NSAIDs) should be used cautiously. Experimental mechanism-based approaches aim to enhance clearance of misfolded uromodulin — e.g., autophagy induction via calorie restriction.

> *"Her gout had been successfully controlled with varying doses of daily allopurinol"* PMID: 25417683
>
> *"Allopurinol has been suggested as a therapeutic agent, but unfortunately this was based on non-randomised uncontrolled trials with small patient numbers."* PMID: 26872483

Finding 9 — Model organisms and genotype-phenotype correlation

Multiple mouse models reproduce the human disease. Umod^C93F and Umod^A227T knock-in lines (C3H background) show gain-of-toxic-function from a uromodulin maturation defect, with ER accumulation/hyperplasia in TAL cells, elevated plasma urea/cystatin, impaired urinary concentration, reduced fractional excretion of uric acid, interstitial fibrosis, inflammation, and tubular atrophy; mutation type and allelic status (homozygous worse) modulate onset and severity. A missense Umod mouse models inherited renal fibrosis with ER stress. Transgenic Tg-Umod^C147W mice are used for therapy testing. Umod/THP knockout mice recapitulate the TAL salt-wasting → hyperuricemia/hypertension axis. Genotype-phenotype correlation is evident in a Galician cohort (37 patients, 15 families): hyperuricemia occurred in 76% of p.C255Y versus 50% of p.Q316P carriers, indicating variant-dependent expressivity. UMOD residues are highly conserved across species.

> *"the Umod mutations cause a gain-of-toxic function due to a maturation defect of the mutant uromodulin leading to a dysfunction of thick ascending limb of Henle's loop (TALH) cells of the kidney"* PMID: 23748428
>
> *"Hyperuricemia was observed in 76% of p.C255Y carriers and 50% of p.Q316P carriers"* PMID: 40300987

Finding 10 — Common *UMOD* promoter variants modulate kidney function (distinct from Mendelian disease)

Distinct from the rare coding mutations that cause ADTKD-UMOD, common non-coding UMOD variants (e.g., promoter rs4293393/rs13333226) are among the top GWAS signals for kidney function and blood pressure. The G allele of rs4293393 associates with higher eGFR and lower CKD risk; rs13333226 associates with ESRD, diastolic blood pressure, and is protective against

type 2 diabetic nephropathy. These common variants act by altering uromodulin production and renal sodium reabsorption in the TAL — mechanistically connected to, but epidemiologically separate from, the Mendelian disease.

> *"G-allele carriers at UMOD rs4293393 exhibited significantly higher eGFR values compared to non-carriers"* PMID: 34593962
>
> *"The G allele of rs13333226 was associated with a decreased risk of nephropathy"* PMID: 21738052

Section-by-Section Report

1. Disease Information

Overview. FJHN Type 1 is a rare, autosomal-dominant, slowly progressive tubulointerstitial kidney disease caused by mutations in **UMOD** (uromodulin / Tamm-Horsfall protein). It is characterized by a triad of dominant inheritance, early-onset hyperuricemia/gout, and progressive CKD leading to ESKD in adulthood. Under current nomenclature it is the UMOD subtype of **autosomal dominant tubulointerstitial kidney disease (ADTKD-UMOD)**.

Key identifiers. | Resource | Identifier | |---|---| | MONDO | MONDO:0008073 | | OMIM | 162000 | | Gene (HGNC) | UMOD, HGNC:12559 | | UniProt | P07911 | | Locus | 16p12.3 | | Classification | ADTKD-UMOD (KDIGO 2015) |

Synonyms / alternative names: Familial juvenile hyperuricemic nephropathy (FJHN); uromodulin-associated kidney disease (UAKD); medullary cystic kidney disease type 2 (MCKD2); glomerulocystic kidney disease (UMOD-related); UMOD-related ADTKD.

Data source type: Information is derived from aggregated disease-level resources (OMIM, Orphanet, KDIGO consensus) and from individual-patient case reports and family cohorts in the primary literature, rather than from a single EHR dataset.

2. Etiology

Causal factors: Primarily **genetic** — heterozygous gain-of-toxic-function mutations in *UMOD*. There is no infectious or primary environmental cause. The disease is fully Mendelian.

Genetic risk factors: The causal variants are germline UMOD coding mutations (predominantly missense affecting conserved cysteines in EGF-like/cysteine-rich/D8C domains). An ultra-rare *SLC8A1* nonsense variant co-segregating with a UMOD deletion was associated with **earlier onset** in one Portuguese family, reviving an old (unconfirmed) FJHN–SLC8A1 linkage hypothesis as a possible modifier PMID: 39216982. Common UMOD promoter variants modulate general-population kidney function but do not cause the Mendelian disease (Finding 10).

Environmental/lifestyle risk factors: No established environmental cause. Precipitating factors that accelerate renal decline or gout include volume depletion, NSAIDs, and fenofibrate (which abruptly raised serum creatinine in an FJHN patient) PMID: 24940667. Dietary purine load and alcohol may aggravate gout non-specifically.

Protective factors: No specific genetic or environmental protective factor is established for the Mendelian disease. In the general population, the UMOD rs4293393 G allele and rs13333226 G allele associate with better kidney outcomes (Finding 10).

Gene–environment interactions: Limited data. The SLC8A1 modifier and precipitant sensitivity (fenofibrate, volume depletion) represent the main documented interactions.

3. Phenotypes

Phenotype	Type	Onset	Frequency / severity	Suggested HPO
Hyperuricemia	Lab abnormality	Childhood–adolescence	Common; variant-dependent (76% in p.C255Y vs 50% in p.Q316P)	HP:0002149 (Hyperuricemia)
Gout	Clinical sign/ symptom	Teenage years–adulthood	Frequent	HP:0001997 (Gout)
Progressive CKD / renal insufficiency	Clinical + lab	3rd–5th decade (variable)	Near-universal, progressive	HP:0012622 (Chronic kidney disease)
End-stage kidney disease	Clinical	3rd–7th decade (median ~42 y in one cohort)	Eventual in most	HP:0003774 (Stage 5 chronic kidney disease)
Impaired urinary concentrating ability	Lab/ physiologic	Early	Common	HP:0003642 (Isosthenuria)
Bland urine sediment (minimal proteinuria/ hematuria)	Lab	Throughout	Characteristic (notably minimal)	HP:0000093 (Proteinuria) — minimal/absent
Renal cysts (corticomedullary)	Imaging/ physical	Variable	Inconstant	HP:0000107 (Renal cyst)
Normal/small kidneys	Imaging	Throughout	Common	HP:0000085-area
Hypertension	Clinical sign	Variable	Occurs	HP:0000822 (Hypertension)

Quality-of-life impact: Progressive CKD imposes dialysis dependency or transplantation, with associated morbidity; recurrent gout attacks cause acute pain and disability. Disease-specific QoL instrument data (EQ-5D/SF-36) specific to ADTKD-UMOD are not available; impact is inferred from the general CKD/ESKD and gout literature.

4. Genetic / Molecular Information

Causal gene: *UMOD* (uromodulin), 16p12.3; HGNC:12559; UniProt P07911.

Pathogenic variants: Predominantly **missense** substitutions, with strong clustering at conserved **cysteine** residues in the EGF-like and cysteine-rich/D8C domains (e.g., C147W, R185H, C223G, C255Y, W258G). Rarer classes include in-frame deletions and a reported large terminal-exon deletion. Variants are **germline**, near-absent in gnomAD, and classified **pathogenic/likely pathogenic** per ACMG/AMP. Functional consequence is a **gain-of-toxic-function** (ER retention of misfolded protein) rather than simple haploinsufficiency, though uromodulin synthesis is also markedly reduced. A dominant-negative component exists: some mutant isoforms escape the ER, reach the plasma membrane, and form extracellular aggregates that impair co-expressed wild-type protein PMID: 22237754.

Modifier genes: A candidate SLC8A1 modifier associated with earlier onset in one family (P 39216982); allelic status (homozygous vs heterozygous) modulates severity in mice.

Epigenetic / chromosomal abnormalities: No disease-specific epigenetic signature or recurrent chromosomal abnormality is established; one large intragenic UMOD deletion is reported.

5. Environmental Information

No environmental toxin, infectious agent, or occupational exposure causes this Mendelian disease. Relevant lifestyle/iatrogenic modifiers: avoid volume depletion, NSAIDs, and fenofibrate (associated with acute renal function decline) PMID: 24940667. Infectious agents: not applicable.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A heterozygous **UMOD missense mutation** (usually altering a conserved cysteine) **results in** a misfolded uromodulin protein with disrupted disulfide bonding / maturation defect.

2. Misfolded uromodulin **leads to** retention in the **endoplasmic reticulum** of TAL epithelial cells (instead of normal trafficking to the apical membrane).
3. ER accumulation **triggers** the **unfolded protein response (IRE1/XBP1 branch)** and **ER stress** (demonstrated: XBP1 splicing, chaperone/PDI up-regulation).
4. Sustained ER stress **results in** altered protein and mitochondrial homeostasis, **defective autophagy**, and (branch A) progressive **tubular epithelial cell death** → interstitial inflammation → **fibrosis** → **progressive CKD → ESKD** (3rd–7th decade).
5. In parallel (branch B), reduced functional uromodulin at the apical membrane **impairs NKCC2-mediated salt reabsorption** in the TAL → salt/water wasting and reduced urinary concentrating ability.
6. TAL salt wasting **leads to** compensatory **up-regulation of proximal-tubule Na-coupled urate transporters** (URAT1/SLC22A12, SLC5A8) and NHE3 (demonstrated in Umod-knockout mice) → increased urate reabsorption.
7. Increased reabsorption **results in** reduced **fractional excretion of uric acid** → **hyperuricemia** → **gout** (early onset).
8. (Minor branch, inferred) Some mutant isoforms escape the ER and form extracellular aggregates exerting a dominant-negative effect on wild-type uromodulin.

Molecular pathways / cellular processes: UPR/ER-stress signaling (IRE1–XBP1), proteostasis/chaperone network, autophagy (impaired; a therapeutic target), apoptosis/cell death, mitochondrial dynamics, interstitial inflammation and fibrosis. **GO term suggestions:** GO:0030968 (endoplasmic reticulum unfolded protein response), GO:0036498 (IRE1-mediated UPR), GO:0034976 (response to ER stress), GO:0006914 (autophagy), GO:0006915 (apoptotic process), GO:0008217 (regulation of blood pressure), ion transport terms for TAL salt handling.

Cell types / anatomy: The thick ascending limb epithelial cell is the primary target. **CL suggestions:** kidney loop of Henle thick ascending limb epithelial cell; proximal tubule epithelial cell (secondary, urate handling). **Subcellular:** endoplasmic reticulum (GO:0005783) is the central compartment; mitochondria (GO:0005739), autophagosome (GO:0005776).

Biochemical abnormalities: Reduced fractional uric-acid excretion; impaired urinary concentration; elevated serum uric acid; in models, elevated renin and allantoin.

7. Anatomical Structures Affected

- **Primary organ:** Kidney (UBERON:0002113), specifically the **thick ascending limb of the loop of Henle** (nephron loop) and distal tubule where inclusions are seen.

- **Secondary/proximal involvement:** Proximal tubule (compensatory urate transport); renal interstitium (fibrosis, inflammatory infiltrate).
- **Body systems:** Renal/urinary system; secondary cardiovascular involvement (hypertension); musculoskeletal (gouty arthritis — first metatarsophalangeal joint classically).
- **Tissue/cell level:** Renal tubular epithelium (epithelial tissue); TAL epithelial cells; interstitial fibroblasts.
- **Subcellular:** Endoplasmic reticulum (bundled/cystic ER on EM), intracytoplasmic uromodulin aggregates.
- **Lateralization:** Bilateral (systemic genetic disease affecting both kidneys).

8. Temporal Development

- **Onset:** Hyperuricemia can appear in childhood; gout commonly in teenage years; CKD becomes clinically apparent in the 3rd–5th decade. Onset pattern is **insidious/chronic**.
- **Progression:** Slowly progressive to ESKD, typically between the 3rd and 7th decades (median ESKD ~41.9 years in one Chinese cohort; variable across variants/families).
- **Course:** Progressive and chronic/lifelong; not relapsing-remitting (gout attacks episodic, superimposed).
- **Critical periods:** Early disease (pre-fibrotic) is the proposed window for proteostasis-based intervention, supported by mouse calorie-restriction data showing greater benefit at an early stage.

9. Inheritance and Population

- **Inheritance:** Autosomal dominant; high but **age-dependent penetrance**; **variable expressivity** (variant-dependent, e.g., C255Y vs Q316P).
- **Sex ratio:** ~1:1 (autosomal).
- **Epidemiology:** No precise prevalence/incidence established; under-diagnosed. ADTKD overall ≈5% of monogenic CKD; UMOD is the most common ADTKD gene; UMOD among the top 8 genes covering >60% of monogenic CKD variants in adults.
- **Founder effects / consanguinity:** Not a prominent feature (dominant disease); recurrent variants (e.g., C255Y, Q316P) reported in regional cohorts (Galicia). De novo mutations occur.
- **Genetic anticipation / germline mosaicism:** Not characteristic.

10. Diagnostics

- **Laboratory:** Serum uric acid (elevated, with low fractional excretion of uric acid — a key discriminator), serum creatinine/eGFR (declining), bland urinalysis (minimal proteinuria/hematuria). Reduced urinary and serum uromodulin correlate with disease severity PMID: 37835820.
- **Imaging:** Renal ultrasound — normal or small kidneys, occasional corticomedullary cysts.
- **Histopathology (biopsy):** Chronic tubulointerstitial fibrosis, tubular atrophy, intracytoplasmic uromodulin aggregates in distal tubule cells; EM shows bundled/cystic ER; IHC confirms uromodulin aggregates.
- **Genetic testing (gold standard):** *UMOD* single-gene sequencing or ADTKD/CKD gene panels (UMOD, MUC1, REN, HNF1B, SEC61A1); WES/WGS useful when panels are negative. Chromosomal microarray can detect large UMOD deletions.
- **Clinical criteria / differential diagnosis:** ADTKD criteria require a pathogenic variant in a defined gene. Differentials: other ADTKD subtypes (MUC1, REN, HNF1B, SEC61A1), ADPKD, nephronophthisis, FSGS (UMOD can mimic), and other causes of juvenile gout (e.g., methylmalonic acidemia, HGPRT deficiency, glycogen storage disease). Low fractional excretion of uric acid plus dominant family history points to UMOD.
- **Screening:** Cascade genetic screening of at-risk relatives once a familial variant is identified.

11. Outcome / Prognosis

- **Natural history:** Progressive CKD to ESKD requiring renal replacement therapy, typically 3rd–7th decade; substantial inter-individual and inter-family variability.
- **Mortality:** Driven by ESKD and cardiovascular comorbidity if untreated; dialysis/transplant markedly improve survival. Disease-specific mortality statistics are sparse.
- **Prognostic factors:** Variant type and domain affected (D8C-domain variants associated with more severe phenotype than GPI-area in one cohort), allelic status, modifier alleles (e.g., SLC8A1), and degree of fibrosis at biopsy.
- **Recovery:** Kidney function loss is not reversible; transplantation is curative for the renal phenotype (no graft recurrence).

12. Treatment

- **Pharmacotherapy (supportive):** Xanthine oxidase inhibitors — **allopurinol** (CHEBI:40279; NCIT C614) and **febuxostat** — for hyperuricemia/gout; colchicine for gout prophylaxis/

flares. Urate-lowering benefit on renal progression is unproven (weak, uncontrolled evidence). Standard nephroprotection (BP control; caution with RAAS agents given salt-wasting tendency).

- **Advanced/experimental:** Autophagy-enhancing proteostasis strategies (calorie restriction in Tg-Umod^C147W mice degrade mutant uromodulin and ameliorate fibrosis). No approved gene, cell, or RNA therapy; these remain conceptual.
- **Surgical/interventional:** Kidney transplantation for ESKD — highly effective, **no disease recurrence** in the graft. Dialysis as needed.
- **Agents to avoid:** Fenofibrate (associated with acute creatinine rise), NSAIDs, volume depletion.
- **NCIT suggestions:** Allopurinol (C614), Febuxostat, Colchicine, Kidney Transplantation, Dialysis.

13. Prevention

- **Primary prevention:** Not possible for the Mendelian genotype; genetic counseling and reproductive options (prenatal/preimplantation genetic testing) can prevent transmission.
- **Secondary prevention:** Cascade genetic screening of relatives; early detection enables early urate control and nephroprotection.
- **Tertiary prevention:** Gout prophylaxis, BP management, avoidance of nephrotoxins, timely transplant planning.
- **Counseling:** Autosomal-dominant counseling (50% transmission risk; variable expressivity and age-dependent penetrance).

14. Other Species / Natural Disease

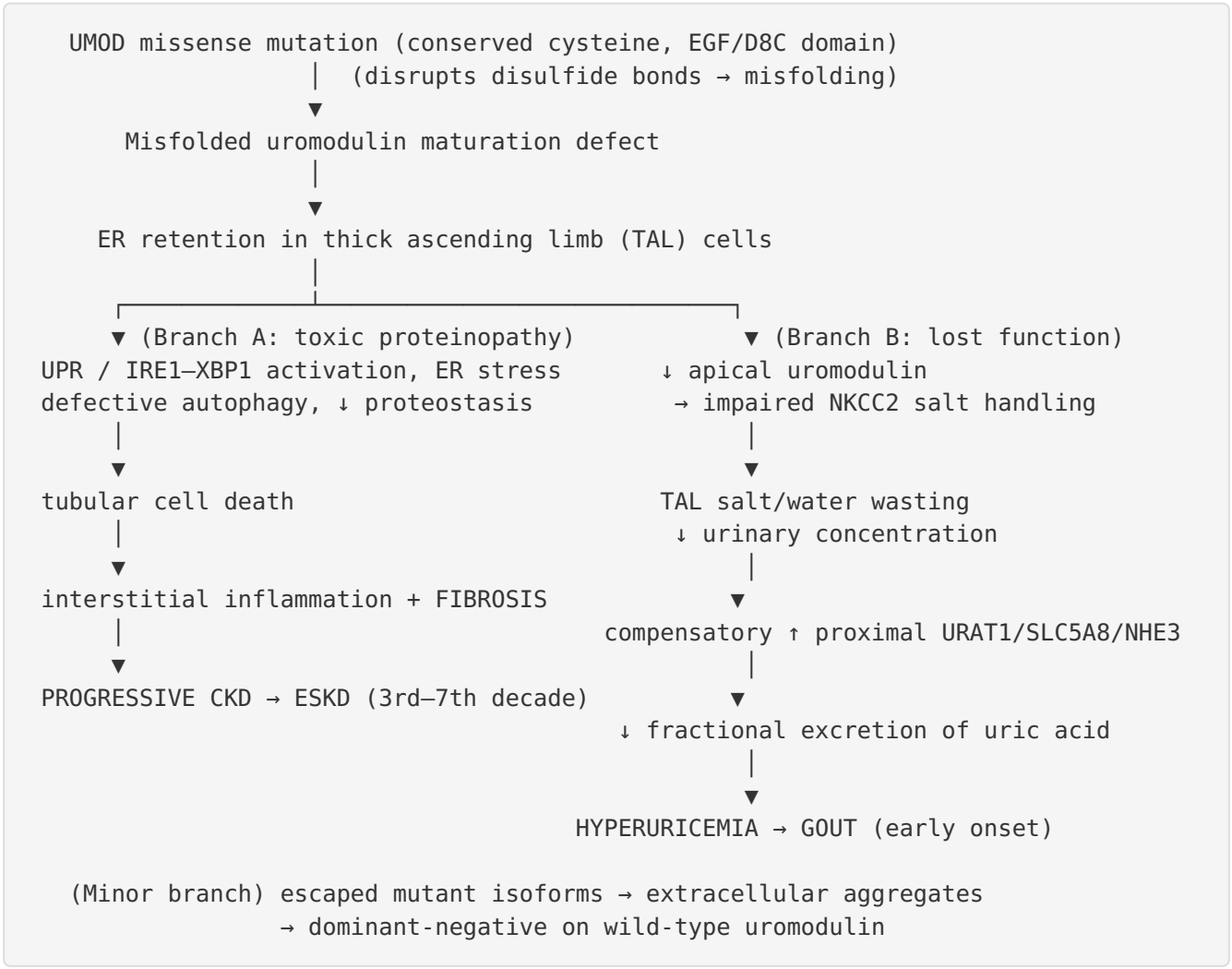
- **Orthologs/taxonomy:** *UMOD* is conserved; mouse *Umod* (NCBI Gene) is the principal ortholog used in modeling. UMOD residues (e.g., V550) are highly conserved across species.
- **Natural disease in other species:** No well-characterized spontaneous animal disease analogue is documented in the reviewed literature; models are engineered (below).

15. Model Organisms

Model	Type	Key phenotype recapitulation	PMID
Umod^C93F, Umod^A227T knock-in (C3H)	Mouse knock-in	ER accumulation in TAL, impaired concentration, reduced FE uric acid, fibrosis, inflammation; severity depends on mutation/allelic status	23748428
Missense Umod mouse	Mouse	Inherited renal fibrosis with ER stress	28325753
Tg-Umod^C147W	Transgenic mouse	Used for therapy testing; calorie restriction reverts/delays phenotype	41632531
Umod/THP knockout	Mouse KO	TAL salt-wasting → compensatory proximal urate reabsorption → hyperuricemia, hypertension	29357410
Immortalized mTAL cells (mutant GFP-UMOD)	Cellular/in vitro	UPR/IRE1 activation, XBP1 splicing, chaperone up-regulation	28437467

Limitations of models: Knockout models capture the salt-handling/urate axis but not the gain-of-toxic-function ER proteinopathy; knock-in/transgenic models better capture ER retention but disease tempo and hyperuricemia penetrance differ from humans.

Mechanistic Model / Interpretation



The two arms converge on the clinical phenotype: the **proteinopathy arm** (Branch A) drives the fibrotic CKD trajectory, while the **loss-of-function/salt-handling arm** (Branch B) drives the hyperuricemia. Upstream, the single rate-limiting lesion is ER retention of misfolded uromodulin; this is why **proteostasis-directed therapy (autophagy enhancement)** is mechanistically attractive — it targets the shared upstream node. The common-variant GWAS biology (Finding 10) is a mechanistically adjacent but epidemiologically separate phenomenon: non-coding variants tune uromodulin *dosage* and TAL sodium reabsorption across the general population, whereas the Mendelian coding mutations produce a toxic misfolded species.

Evidence Base

PMID	Role in report
25738250	KDIGO consensus establishing ADTKD classification and UMOD as cause
26872483	Causal gene/locus, clinical triad; weak allopurinol evidence
28437467	UPR/IRE1–XBP1 activation by mutant uromodulin (in vitro)
41632531	Gain-of-toxic-function summary; calorie-restriction therapy in mice
29357410	TAL salt-wasting → compensatory urate reabsorption (KO mouse)
23748428	Knock-in mouse models; reduced FE uric acid; genotype effects
31422399	Diagnostic histopathology (intracytoplasmic uromodulin aggregates)
31488840	Epidemiology: ADTKD ~5% of monogenic CKD
38958599	UMOD among top monogenic CKD genes in adults
21071970	Age of ESKD, early hyperuricemia/gout
25417683	Renal under-excretion basis of hyperuricemia; allopurinol use
32815518 / 33196145	Cryo-EM uromodulin filament / ZP-module assembly
40300987	Variant-dependent expressivity (C255Y vs Q316P)
29569962	Chinese cohort; median onset ~26.5 y, ESKD ~41.9 y; D8C vs GPI severity
22237754	Dominant-negative extracellular aggregates (escaped mutant isoforms)
24940667	Fenofibrate-associated acute renal decline (agent to avoid)
39216982	Large UMOD deletion; SLC8A1 candidate modifier (earlier onset)

PMID	Role in report
34593962 / 21738052 / 27938332 / 22592667	Common UMOD variant GWAS biology (distinct from Mendelian disease)
28325753	Missense Umod mouse — inherited fibrosis with ER stress
30099615	ER stress in monogenic kidney disease; MANF/CRELD2 biomarkers
37835820	Uromodulin as kidney-disease biomarker

Evidence-type mix: Human clinical (cohorts, case reports, KDIGO consensus), model organism (knock-in/KO/transgenic mice), in vitro (mTAL cells), and computational/structural (cryo-EM, AlphaFold). Convergent across all four source types for the central ER-retention mechanism.

Limitations and Knowledge Gaps

- **Epidemiology:** No precise population prevalence/incidence; the disease is under-diagnosed. Reported figures are relative (fraction of monogenic CKD) rather than absolute.
- **Therapeutics:** No disease-modifying therapy is proven in humans. Allopurinol's effect on renal progression rests on small, uncontrolled studies. Calorie-restriction/autophagy data are preclinical (mouse only).
- **Genotype–phenotype:** Variant-dependent expressivity is documented but incompletely predictive; modifiers (e.g., SLC8A1) are hypothesis-generating from single families and unconfirmed.
- **Mechanistic uncertainty:** The relative contributions of the toxic-gain (fibrosis) arm versus loss-of-function (urate) arm to the final clinical trajectory are not precisely quantified in humans. The pathogenicity mechanism of the large terminal-exon UMOD deletion (which spares core domains) is unexplained.
- **QoL / outcomes:** No disease-specific quality-of-life, survival, or natural-history registry data (EQ-5D/SF-36) are available; prognosis is extrapolated from the general CKD/ESKD literature.
- **Human proteostasis therapy:** Whether autophagy enhancement translates to humans, and at what disease stage, is untested.

Proposed Follow-up Experiments / Actions

1. **Clinical translation of proteostasis therapy:** Design early-phase trials or dietary/pharmacologic autophagy-induction studies (e.g., intermittent fasting, mTOR modulation) in genetically confirmed early-stage ADTKD-UMOD patients, using serum/urinary uromodulin and eGFR slope as endpoints.
2. **Natural-history registry:** Establish a multinational ADTKD-UMOD registry capturing variant, age of gout/CKD onset, ESKD age, and QoL to define prevalence, penetrance curves, and genotype-stratified prognosis.
3. **Validate the SLC8A1 modifier:** Test SLC8A1 co-segregation and onset effect in additional UMOD families and model it functionally.
4. **Mechanism quantification:** Use TAL organoids / iPSC-derived tubular cells carrying patient cysteine variants to quantify ER-stress magnitude, autophagic flux, and dominant-negative extracellular aggregation across variants.
5. **Biomarker development:** Validate ER-stress biomarkers (MANF, CRELD2) and serum/urinary uromodulin as non-invasive diagnostic/progression markers specific to ADTKD-UMOD.
6. **Deletion pathogenicity:** Functionally dissect the large terminal-exon UMOD deletion to clarify how a variant sparing core domains causes disease.

Report compiled from 10 confirmed findings and 37 reviewed papers over 5 investigation iterations. Causal gene: UMOD (uromodulin). Disease: FJHN Type 1 = ADTKD-UMOD (MONDO:0008073; OMIM 162000).
